How to Identify Tremor Dominant and Postural Instability/Gait Difficulty Groups With the Movement Disorder Society Unified Parkinson's Disease Rating Scale: Comparison With the Unified Parkinson's Disease Rating Scale

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ABSTRACT

Background: Formulas were developed to define *tremor dominant* (TD) and *postural instability/gait difficulty* (PIGD) phenotypes of Parkinson's Disease (PD) using the Movement Disorder Society Unified Parkinson's Disease Rating Scale (MDS-UPDRS). TD and PIGD designations, based on the original Unified Parkinson's Disease Rating Scale (UPDRS), provided useful designations for classifying different phenotypes of PD. With the advent of the MDS-UPDRS, a valid set of calculations for these phenotypes is needed.

Methods: UPDRS and MDS-UPDRS scores were collected on 877 PD patients. TD/PIGD scores were calculated using the UPDRS formula for all patients. Comparable TD and PIGD items from the MDS-UPDRS were used to calculate new ratios. Data were analyzed using receiver operating characteristic models.

Results: The new MDS-UPDRS TD/PIGD ratios accounted for a significant area under the curve compared with the UPDRS classification. Optimal sensitivity and specificity were obtained with MDS-UPDRS cutoff scores of $\geq$1.15 for TD classification and $\leq$0.90 for PIGD.

Conclusions: The development of comparable and valid PIGD and TD scores from the MDS-UPDRS provides a clear method for clinicians and researchers to transition from the original UPDRS to the new MDS-UPDRS in categorizing patients with different clinical phenotypes. © 2013 *Movement* Disorder Society

Key Words: Parkinson's disease; Rating scales; Movement Disorder Society Unified Parkinson's Disease Rating Scale; MDS-UPDRS

Different clinical phenotypes of Parkinson's disease (PD) have long been recognized, and the prototypic divisions of tremor dominant (TD) versus postural instability/gait difficulty (PIGD) parkinsonism have been utilized in clinical studies, risk factor analyses, and genetic surveys.[1–9] Differentiation of TD from PIGD motor impairment in PD has been anchored in defined and widely accepted algorithms using the Unified Parkinson's Disease Rating Scale (UPDRS).[1] In 2008, the Movement Disorder Society (MDS) released a revision to the UPDRS, called the MDS-UPDRS.[10] In the clinimetric program to establish this new scale, the reliability and factor structure of the MDS-UPDRS were validated using a sample of over 800 PD patients evaluated with both the UPDRS and the MDS-UPDRS. In this study, we identified the group of patients that fulfilled criteria for the UPDRS-based TD and PIGD subtypes of PD and developed algorithms for the same designation based on MDS-UPDRS scores.

Patients and Methods

The MDS-UPDRS validation sample focused on the recruitment of 877 patients with PD of diverse race/ethnicity representations (78% non-Latino Caucasians). The average PD duration ($\pm$ standard deviation) was 8.3$\pm$6.7 years with varying levels of PD severity as measured by Hoehn and Yahr staging (stage I, 7%; stage II, 53%; stage III, 20%; stage IV, 12%; stage V, 6%). On a single visit, both the UPDRS and the MDS-UPDRS were administered.[7]

UPDRS items for TD and PIGD designations were used to calculate mean TD and PIGD scores.[1] Following the original classification methods, the ratio of the mean UPDRS tremor scores (8 items) to the mean UPDRS PIGD scores (5 items) was used to define TD patients (ratio $\geq$1.5), PIGD patients (ratio $\leq$1), and indeterminate patients (ratios $>$1.0 and $<$1.5) (Table 1, left column). In addition, patients who had a positive mean in the numerator and a zero in the

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TABLE 1. Items used for tremor dominant and postural instability/gait difficulty calculations[a]

UPDRS	MDS-UPDRS
Tremor score	Tremor score
Part II	Part II
2.16. Tremor	2.10. Tremor
Part III	Part III
3.20. Rest tremor face	3.15. Postural tremor RUE
3.20. Rest tremor RUE	3.15. Postural tremor LUE
3.20. Rest tremor LUE	3.16. Kinetic tremor RUE
3.20. Rest tremor RLE	3.16. Kinetic tremor LUE
3.20. Rest tremor LLE	3.17. Rest tremor RUE
3.21. Action tremor RUE	3.17. Rest tremor LUE
3.21. Action tremor RUE	3.17. Rest tremor RLE
	3.17. Rest tremor LLE
	3.17. Rest tremor lip/jaw
	3.18. Rest constancy
PIGD score	PIGD score
Part II	Part II
2.13. Falling	2.12. Walking and balance
2.14. Freezing	2.13. Freezing
2.15. Walking	
Part III	Part III
3.29. Gait	3.10. Gait
3.30. Postural stability	3.11. Freezing of gait
	3.12. Postural stability

[a]In the UPDRS calculations (first column), the mean values of the tremor scores (0–4 for each item)/the mean values for the PIGD scores (0–4 for each item) divides clinical phenotypes into TD (ratio ³1.5) and PIGD (ratio £1) categories. In the second column, the comparable items tested for an MDS-UPDRS algorithm are shown. To calculate the MDS-UPDRS TD/PIGD score, the mean of MDS-UPDRS items 2.10, 3.15a, 3.15b, 3.16a, 3.16b, 3.17a, 3.17b, 3.17c, 3.17d, 3.17e, and 3.18 is divided by the mean of MDS-UPDRS items 2.12, 2.13, 3.10, 3.11, and 3.12. If the resultant ratio is ≥1.15. then the patient is classified with TD. If the ratio is ≤0.90, then the patient is classified with PIGD. If the ratio is between 0.90 and 1.15, then the patient is classified as indeterminate.

denominator were classified as TD; patients with a zero in the numerator and a positive mean in the denominator were classified as PIGD, and patients with zeros in both the numerator and denominator were classified as indeterminate (Michael McDermot, PhD, personal communication, April 6, 2012).

We then selected MDS-UPDRS items comparable to the TD and PIGD items on the UPDRS (11 items for tremor and 5 items for PIGD) (Table 1, right column) and developed PIGD and TD ratio scores for all patients. Using the UPDRS TD and PIGD classifications as the comparator, we applied receiver operating characteristics (ROC) analyses to MDS-UPDRS ratios to determine the area under the curve (AUC) and ideal cutoff scores for sensitive and specific group identification.

Results

Complete UPDRS and MDS-UPDRS data sets for TD and PIGD items were available for 847 PD patients. Of this cohort, 203 patients met the criteria for TD (24%), 543 met the criteria for PIGD (64%), and 101 were classified as indeterminate (12%) based on the UPDRS definitions. ROC analyses for the newly created MDS-UPDRS TD/PIGD ratios were significant. Using the UPDRS TD classification, the new MDS-UPDRS TD/PIGD ratio had an AUC of 94% (0.939; $P<0.0005$; 95% confidence interval, 0.919–0.959). Significant results also were obtained when we used the UPDRS PIGD classification, with a new MDS-UPDRS ratio that demonstrated an AUC of 92% (0.923; $P<0.0005$; 95% confidence interval, 0.900–0.946). Examination of the ROC curve revealed optimum cutoff levels for the MDS-UPDRS TD/PIGD ratio of ≥1.15 for TD classification (sensitivity=0.89; sensitivity=0.90; positive predictive value=0.96; negative predictive value=0.95) and ≤0.90 for PIGD classification (sensitivity=0.91; sensitivity=0.85 positive predictive value=0.97; negative predictive value=0.92).

Discussion

The heterogenous presentation of PD includes differing age at onset, side predominance, and a variety of clinical manifestations.[11] The TD and PIGD phenotypes have been linked to differing associated features, genetic patterns, and risk for developing complications, including cognitive decline.[1–9] Although the formulas used to define TD and PIGD subtypes were derived empirically, their definitions have been used continually in research efforts.[1] The method was based on a simple derivative of items from the UPDRS Activities of Daily Living portion (part II) and the Motor Examination portion (part III), which were pertinent to either tremor or midline gait and balance issues and their mean ratio. This approach to developing classification algorithms has been contrasted with other data-driven approaches, such as cluster analyses.[12] However, the aim of this study was to provide an empiric method to retain a time-honored classification system using a new scale, namely, the MDS-UPDRS. Whereas these 2 subgroups identify prototypic cases of 2 distinct phenotypes, a large group of patients with PD fall between these 2 subtypes and have relatively balanced involvement of tremor and gait/balance impairments or disabilities.

The MDS-UPDRS was developed in response to an MDS-directed critique of the UPDRS.[13] In this critique, the original scale was lauded for its utility in clinical care and research efforts, but its weaknesses included a number of ambiguities and the lack of specific instructions for the application of the scale. Furthermore, the original scale mixed 0 to 5 ratings with yes/no responses, making calculations difficult across the entire scale. Finally, the UPDRS insufficiently assessed a number of nonmotor elements implicit to PD or regularly encountered in relation to treatment.

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Based on this critique, the MDS initiated a revision process that resulted in the development and full clinimetric testing of the MDS-UPDRS (http://www.movementdisorders.org/publications/rating_scales/). Although the scale was developed and tested originally in English, official, clinimetrically tested non-English versions of the MDS-UPDRS now exist in Italian, Estonian, and Spanish. Other language versions are being developed, and the current program includes development teams for French, German, Russian, Hebrew, Japanese, Chinese, Korean, Thai, and Slovakian. Other language teams can organize through the MDS (contact cgoetz@rush.edu).

To facilitate the transition from the UPDRS to the MDS-UPDRS, a clear set of formulas to define TD and PIGD forms of PD based on MDS-UPDRS scores is essential. Previously, we provided recommendations on the methods to transfer total scores on part III of the UPDRS to comparable total scores on the MDS-UPDRS.[14] That analysis did not focus on individual item values, and this current analysis allows this more specific score transfer from the older scale to the new revision to identify 2 archetypal phenotypes of PD. Our results demonstrate that TD and PIGD scores can be calculated from the MDS-UPDRS and clearly identify the same patient groups defined by the original UPDRS. The conversion formulas are based on a large data set that covered all Hoehn and Yahr stages, a wide range of disease durations, and included patients on all levels of medications from no drugs to multiple medications, including levodopa. The majority of patients, however, had mild to moderate PD severity, and a broader severity range might improve classification in the group of patients with Hoehn and Yahr stage IV/V PD. The sample demographics in our study may account for the differential proportion of UPDRS-based TD and PIGD cases (24% and 64%, respectively) versus the proportions of these phenotypes in the original sample on which the TD/PIGD algorithms were based (55% and 29%, respectively).[1] Regardless, in this sample, the MDS-UPDRS TD and PIGD items account for over 90% of the shared variance with UPDRS TD and PIGD scores. The formulae used in the UPDRS calibrates with ease to the MDS-UPDRS, allowing clinicians and researchers to follow the same formerly developed, time-honored calculations. These results provide a validated method for calibrating UPDRS TD/PIGD scores to MDS-UPDRS usage and will be useful as the MDS-UPDRS replaces the UPDRS in clinical and research trials. ■

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